

Using Ratio Tables

Grade 6 Mathematics

Name: _____ Date: _____

Learning Objective

We will create ratio tables to organize equivalent ratios and use them to find missing values in real-world problems.

Do Now

Directions: Answer all questions. Show your thinking and be ready to share your reasoning.

1. Circle all the ratios below that represent the same relationship as 3:5

- 6:10
- 5:3
- 15:25
- 9:12
- 12:20

Explain your thinking: How did you decide which ratios were equivalent?

2. Jamal's Trail Mix Recipe:

- 2 cups of nuts
- 3 cups of dried fruit

a) If Jamal uses 6 cups of nuts, how many cups of dried fruit does he need? _____

b) Circle the statement that is TRUE:

- The ratio of nuts to fruit is always 2 to 3, no matter how much trail mix Jamal makes
- The ratio changes depending on how much trail mix Jamal makes
- The ratio is only 2 to 3 when he uses exactly those amounts

c) Why is your answer to part (b) true? Explain in your own words.

3. Think About It:

Marcus says: *"If I double both numbers in a ratio, I'm making a new, different ratio."*

Do you agree or disagree? Use an example to support your answer.

Introduction

Today we're learning how to use tables to organize ratios. Tables help us see patterns and find missing numbers quickly. This will make solving ratio problems much easier!

Notes:

Example: Fruit Punch Recipe

The school cafeteria makes fruit punch using 2 cups of orange juice for every 3 cups of pineapple juice.

Ratio Table:

Orange Juice (cups)	Pineapple Juice (cups)
2	3
4	6
6	9
8	12
10	15

Sample Problem: The cafeteria needs to make punch with 15 cups of pineapple juice. How much orange juice do they need?

Answer: 10 cups of orange juice

Notes:

Common Questions

Q: Do I always have to start with the original ratio in my table?

A: Yes! Always put the original ratio in the first row. This helps you see the pattern clearly.

Q: How do I know what number to multiply by?

A: Look at what number you're trying to find. Ask yourself: "What do I multiply the original by to get close to this number?"

Q: What if the number I need isn't in my table?

A: You can always add more rows! Just keep multiplying by the next whole number (or find the right multiplier).

Q: Do I have to fill in every row?

A: No. You only need enough rows to solve the problem. But more rows can help you see the pattern better.

Q: Can I multiply by any number?

A: Yes! You can multiply by 2, 3, 4, 5, or even 10. Just multiply BOTH parts of the ratio by the same number.

Q: What if my table gets really big with large numbers?

A: That's okay! Sometimes we need big numbers to solve the problem. The pattern stays the same no matter how big the numbers get.

Guided Practice

Problem: A paint mixture uses 4 parts blue paint to 7 parts yellow paint to make green.

Create a ratio table:

Blue Paint (parts)	Yellow Paint (parts)
4	7
8	
12	
16	
20	

Questions:

1. If you have 16 parts blue paint, how many parts yellow paint do you need?

Answer: _____

2. If you have 28 parts yellow paint, how many parts blue paint do you need?

Answer: _____

Notes:

Independent Practice

Problem: A recipe for lemonade uses 4 lemons for every 6 cups of water.

1. Create a ratio table showing at least 5 rows:

Lemons	Water (cups)

2.
How many lemons are needed for 18 cups of water?

Answer: _____

3. If you use 12 lemons, how many cups of water do you need?

Answer: _____

Notes:

Challenge Problems (Optional)

Challenge 1: Could you make lemonade with 15 cups of water using this recipe? Explain why or why not.

Challenge 2: A different lemonade recipe uses 5 lemons for every 8 cups of water. If you want to make lemonade with 40 cups of water, which recipe uses fewer lemons? Show your work using ratio tables.

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Exit Ticket

A baker uses 2 eggs for every 3 cups of flour.

1. Complete this ratio table:

Eggs	Flour (cups)
2	3
4	
	9
8	
	15

2. If the baker uses 12 eggs, how many cups of flour are needed? _____

3. Explain how you used the table to find your answer.